

Dr.-Ing. Suresh Gopalkrishna

Simulation Engineer

Address: München, Germany

Email: suresh.gopalkrishna@icloud.com

LinkedIn: linkedin.com/in/sureshgopalkrishna/



Work History

06/2023 – present

Research Consultant

Otto von Guericke University Magdeburg – ovgu.de

Magdeburg, Germany

- Applied an inverse heat conduction model to 50+ experimental cases, using Python for transient heat transfer of metal quenching process.
- Developed and validated a Physics-Informed Neural Network (PINN) model, achieving a 2.7% mean relative heat flux error on synthetic data.
- Applied the PINN model to experimental AA6082 IR data, recovering a peak heat flux of 10.9 MW/m² with a temperature RMSE of 1.73K.
- Published 2+ peer reviewed articles and collaborated with research partners on computational modeling, experimental validation, and data-driven thermal analysis.

12/2018 – 05/2023

Research Associate

Otto von Guericke University Magdeburg – ovgu.de

Magdeburg, Germany

- Developed and validated an in-house 2D/3D FEM solver in MATLAB for inverse problems, enabling prediction of transient heat flux distributions.
- Executed 500+ FEM simulations in MATLAB and COMSOL to quantify heat flux and thermal stresses during hot metal cooling.
- Conducted 400+ thermal experiments using infrared thermography at 200 fps, supporting numerical model development and validation.
- Led 7 master's research projects in heat transfer, cooling, and numerical modeling, contributing to 2+ collaborative thermal engineering projects.
- Published and presented 5+ technical studies on industrial thermal processes and intensive cooling of hot metals.

03/2016 – 09/2016

Mechanical Engineer

Narsipur Auto Components Private Limited – narasipur.com

Bengaluru, India (A supplier to TVS Motors India)

- Reduced product failure rates by 90%+ through SPC, 8D, Kaizen, and Six Sigma.
- Received Best Engineer recognition twice for process improvement across 2+ workflows and ISO-compliant quality implementation.

07/2014 – 08/2014

Internship – Helicopter Engine Division

Hindustan Aeronautics Limited, Engine Division – hal-india.co.in

Bengaluru, India

- Analyzed aircraft engine overhaul processes for Rolls-Royce and turboprop engines through technical discussions with in-house.

Education

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|-------------------|--|
| 04/2019 – 10/2022 | PhD, Thermal Engineering
Otto von Guericke University Magdeburg
Magdeburg, Germany
Graduated Magna Cum Laude |
| 10/2016 – 10/2018 | Master of Science, Chemical and Energy Engineering
Otto von Guericke University Magdeburg
Magdeburg, Germany
Graduated Good |
| 09/2011 – 07/2015 | Bachelor of Engineering, Mechanical Engineering
AMC Engineering College
Bengaluru, India
Graduated First Class with Distinction |

Additional Skills

Languages

- Kannada (Mother Tongue), English (C2), German (B2)

IT-Skills

- **Advanced Knowledge:** ANSYS, MATLAB, COMSOL, LaTeX, FEM.
- **Intermediate:** Python, NumPy, SciPy, Pandas, Matplotlib, Scikit-learn, PyTorch, Keras, TensorFlow, Machine Learning, Deep Learning, PINNs, LLMs, Generative AI, CFD, Solid Edge

Certifications

- **AI & Machine Learning:**
 - Supervised Machine Learning: Regression & Classification, Advanced Learning Algorithms – DeepLearning.AI
 - Python for Data Science, AI & Development – IBM,
 - AI for Mechanical Engineering Specialization – University of Michigan
- **Simulation & CFD:** Applied Computational Fluid Dynamics – Siemens
- **Programming:** Introduction to Programming & Python - Kaggle

August 18, 2026
Suresh Gopalkrishna